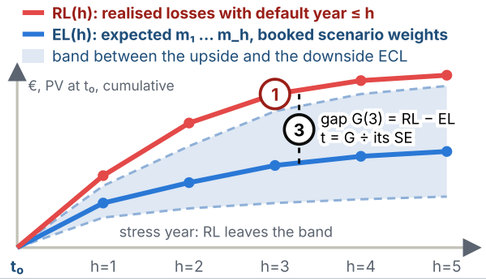
Top-Down Multiyear Backtest of IFRS 9 ECL

Residential mortgages · the ECL booked at a reporting date vs. the loss later realised on the same loans · ECL level only, no component analysis

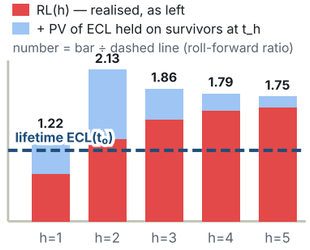
Main idea. At a reporting date t_o the bank booked an ECL for every loan — a forecast of what those loans will lose, in today's money. The backtest keeps exactly those loans, waits h = 1 … 5 years, measures what they actually lost, discounts it back to t_o the same way and compares the two sums. One cohort answers ① **adequacy** (was the provision enough?), five cohorts answer ② **bias** (wrong in the same direction every time?), a paired t-test answers ③ **fit** (is the gap more than noise?). Loans still on book at t_h are either left out on both sides (**truncation test**) or carried at the ECL they hold at t_h (**roll-forward test**) — panel A shows both.

A The comparison — and the two ways to treat loans still on book

Truncation test — RL(h) vs EL(h); survivors left out on both sides



Roll-forward test — survivors carry ECL(t_h)



THE TWO TESTS, DEFINED

Truncation (step 5): $RL(h) \lessgtr EL(h) = \Sigma_i \Sigma_{k\le h}\, m_{i,k}$ — both sides stop at year h: a survivor adds 0 to RL and only $m_1 \ldots m_h$ to EL. Tests the first h years of the term structure, nothing beyond.

Roll-forward (step 6): $\Sigma_i \Sigma_k\, m_{i,k} \lessgtr RL(h) + \Sigma_{\text{survivors}}\, (1+r_i)^{-h}\cdot ECL_i(t_h)$ — lifetime ECL at t_o against realised loss plus the *shadow* lifetime ECL each survivor holds at t_h (best estimate of what is still to come). Tests the whole provision at every h; but the right side is the engine's own view — an optimistic engine passes its own roll-forward (h = 1 above) — so read it next to the truncation ratio, never alone.

- ① **Adequacy — was the provision enough?** One cohort, one horizon: RL(h) against EL(h). The band is the bank's own plausible range at t_o (upside-scenario ECL … downside-scenario ECL). Inside = adequate; above = worse than the stress case = under-provision (amber, red beyond 125 % of the downside); below = over-provision.
- ② **Bias — wrong in the same direction every time?** One cohort sees one economy; one result can be luck. The 15 ratios ρ = RL/EL of all cohorts form the triangle. One red diagonal = one bad year. Red along rows, down columns and on both sides of it = bias (ECL systematically too low; blue = too high).
- ③ **Fit — is the gap more than noise?** G = RL − EL is a sum of loan-level differences; ÷ its standard error = the paired t (panel E). Valid against the hindsight ECL only; $|t| > 2$ = mis-fit beyond chance even knowing the economy.

② every t_o → the triangle

	h1	h2	h3	h4	h5
2021	+	+	+	+	+
2022	+	+	+	+	
2023	−	−	−		
2024	−	−			
2025	−				

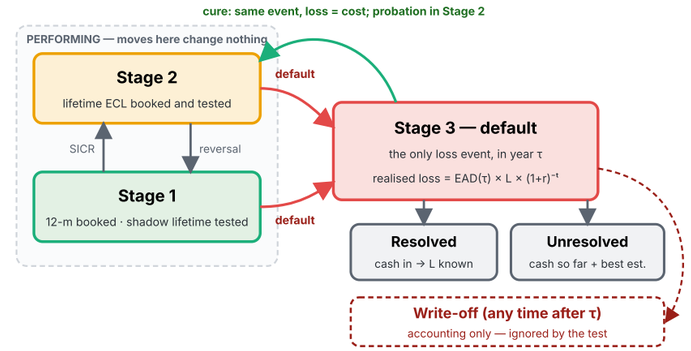
cell = ρ(h) · red > 1 · blue < 1 · grey = not yet

Symbols. i = a loan · $m_{i,k}$ = expected loss of loan i for development year k, exactly as the ECL engine computed it at t_o ($m_{i,1}$ = its 12-month ECL, $\Sigma_k\, m_{i,k}$ = its lifetime ECL) · $RL(h) = \Sigma_{i:\,\tau_i \le h}\, (1+r_i)^{-\tau_i}\cdot \text{EAD}_i(\tau_i)\cdot L_i$: τ_i year of default, r_i EIR, $\text{EAD}_i(\tau_i)$ balance at default, L_i realised LGD (panel F) · $\text{ECL}_i(t_h)$ = shadow lifetime ECL at t_h.

B Procedure

1. **Define the test** — Reference dates 31 Dec 2021 … 2025, cut-off 31 Dec 2026, portfolio. Same conventions both sides: default = 90 dpd / UTP; PV at t_o at the loan's EIR; a loss is dated by its default year.
2. **Freeze one cohort per t_o** — Every loan on book with stage, EAD, EIR, booked ECL, upside / downside ECL and expected loss per year $m_1 \ldots m_L$. Never restated.
3. **Two populations** — (a) Stage 2 + 3 at their booked lifetime ECL; (b) whole book, Stage 1 at its shadow lifetime ECL. Booked ECL as is at h = 1.
4. **Realised loss per loan** — First default after t_o (year τ), EAD then, cash discounted to τ → L; loss = $(1+r)^{-\tau}\cdot\text{EAD}\cdot L$. Cure = cost only; unresolved = cash so far + Stage 3 provision; Stage 3 at t_o: EAD − PV(cash after t_o).
5. **Truncation test, per horizon** — EL(h) = $\Sigma m_1 \ldots m_h$ vs RL(h), h = 1 … H: ratio ρ, gap G, defaults, unresolved share (panel A left).
6. **Roll-forward test** — Lifetime ECL at t_o vs RL(h) + PV of the shadow lifetime ECL held on survivors at t_h (panel A right). Report next to step 5.
7. **Hindsight ECL** — Same engine, same loans, realised macro path. RL − hindsight = the engine's fit; hindsight − booked = what the scenarios missed.
8. **Triangle** — Rows t_o, columns h, diagonals = calendar years. Same sign along rows and columns, on both sides of the stress diagonal = bias.
9. **Grade and escalate** — Green inside the upside–downside band; amber ≤ 25 % outside or unresolved > 25 %; red beyond. Amber → owner explains; red → committee.
10. **Paired t-test** — Per cell: d_i = realised − expected per loan, t = mean(d)/[sd(d)/√n]. Valid with hindsight ECL as expected, indicative with booked (panel E).
11. **Data checks, repeat yearly** — Default definition stable; forbearance, sales and unresolved shares reported per cell. Full triangle every year-end.

C Stage migrations and write-off



Moves between Stage 1 and Stage 2 after t_o

- EL(h) and RL(h) do not change: stage and expected loss per year are frozen at t_o; a loan that moves 1→2 or 2→1 and never defaults has RL = 0.
- They matter in one place only: the roll-forward adds each survivor's *shadow* lifetime ECL at t_h, so a 2→1 move cannot flatter the test.

Cure and re-default

- Cure = same default event; realised loss = workout cost K only. The ECL already prices the chance of a cure, so the expected side is cure-adjusted too.
- Re-default inside probation = same event (cash flows continue); after probation = a second default event of the same loan, counted again — the unit of RL is the default event.

Write-off is not relevant to this test, because

- (a) it is an accounting entry, not a default — the loss is recognised at τ and dated there;
- (b) its timing is policy-driven, 1–4 years after default, so it would misdate and understate losses;
- (c) cash can still arrive after it — only cash and costs enter L, whenever they occur.

The write-off date is therefore not used anywhere on this page.

D Every combination: stage at t_o × what happened by t_h → expected loss (EL) and realised loss (RL)

↓ stage at t_o sets EL · → end state at t_h sets RL	**Still on book, no default** (any 1↔2 moves) — the only column where the two tests differ	**Left without default** prepaid · matured	**Sold / securitised**	**Defaulted in year τ ≤ h, resolved** collateral sold, cash in — written off or not	**Defaulted in year τ ≤ h, cured**	**Defaulted in year τ ≤ h, unresolved at cut-off**
Stage 1 at t_o EL = $m_1+\ldots+m_h$, **shadow** lifetime (whole-book test); booked test at h=1: EL = m_1	EL: Σm as in row RL: 0 (truncation) · $(1+r)^{-h}$·shadow ECL at t_h (roll-forward)	EL: Σm (prepayment is already priced in m) RL: 0	EL: Σm RL: carrying amount − PV(sale price) report share sold	EL: Σm — the booked 12-m ECL never covered τ ≥ 2 RL: $(1+r)^{-\tau}\cdot$EAD(τ)·L, L from all cash incl. post-write-off	EL: Σm (cure chance priced in) RL: $(1+r)^{-\tau}\cdot K \approx$ 1 % of EAD	EL: Σm RL: $(1+r)^{-\tau}\cdot\text{EAD}(\tau)\cdot\hat{L}$, $\hat{L}$ = cash so far + Stage 3 provision at cut-off flag; amber if > 25 % of RL
Stage 2 at t_o EL = $m_1+\ldots+m_h$, **booked** lifetime ECL	EL: Σm RL: 0 (truncation) · $(1+r)^{-h}$·shadow ECL at t_h (roll-forward), even after a 2→1 move	EL: Σm RL: 0	EL: Σm RL: carrying amount − PV(sale price) report share sold	EL: Σm — the case IFRS 9 intends RL: $(1+r)^{-\tau}\cdot$EAD(τ)·L	EL: Σm RL: $(1+r)^{-\tau}\cdot K$	EL: Σm RL: $(1+r)^{-\tau}\cdot$EAD(τ)·$\hat{L}$ flag unresolved
Stage 3 at t_o **(already defaulted)** EL = EAD × in-default LGD, all in h = 1; columns refer to the *open* default	n/a (performing at t_h = cured)	n/a	EL: EAD·LGD^{S3} RL: EAD(t_o) − PV$_{t_o}$(sale price) report NPL sales	EL: EAD·LGD^{S3} RL: EAD(t_o) − PV$_{t_o}$(all cash after t_o), booked at h=1	EL: EAD·LGD^{S3} (cure chance priced in while the cure window is open) RL: K at h=1 a later re-default is a new event of the cured loan	EL: EAD·LGD^{S3} RL: EAD(t_o) − PV(cash so far) − PV(best-estimate remaining) flag unresolved

Both tests read off this matrix. Truncation: EL = $m_1 \ldots m_h$, RL as shown. Roll-forward: EL = the full lifetime $\Sigma_k\, m$, and in the "still on book" column RL = the survivor's shadow lifetime ECL at t_h discounted to t_o; all other columns identical. **All transitions covered?** Yes, by construction: every loan is in exactly one row (its stage at t_o) and one column at t_h (six exclusive, exhaustive end states); whatever happened in between only decides the column, so the cells sum to EL(h) and RL(h) exactly once. Conventions: second default after probation = second event; partial prepayment ≠ exit; EIR frozen at t_o; sold while in default = Stage 3 × sold; forbearance without default = still on book.

E The t-test (step 10): how to use it on ECL and how to read it

PAIRED T-TEST, ONE PER COHORT × HORIZON CELL

d_i = realised loss$_i(h)$ − expected loss$_i(h)$ for every loan i of the cohort; $t = \bar{d} / [\, s_d / \sqrt{n}\,]$, df = n − 1

H_0: mean difference = 0; one-sided alternative "realised > expected" = under-provision. $\Sigma d_i = G(h)$: the test is the gap divided by its loan-level standard error. Loans that never default enter with $d_i = -$expected.

- **On the hindsight ECL — valid.** With the economy held fixed, loan-level differences are independent; t tests whether the engine is biased for these loans. $|t| > 2 \approx$ 5 % two-sided; with 15 cells read the pattern, not one cell.
- **On the booked ECL — indicative only.** All loans share one macro surprise, which the standard error ignores, so $|t|$ is inflated in stress years; adequacy on the booked basis is graded by the band (step 9).
- **Not** on the 15 cell ratios against 1: the cells overlap in loans and years; the effective sample is one or two.

t on **booked** ECL — indicative	h1	h2	h3	h4	h5
2021 (n = 16,732)	+6.7	+10.8	+12.0	+12.4	+12.5
2022 (n = 15,554)	+13.1	+13.7	+13.8	+13.8	n/a
2023 (n = 14,477)	−7.2	−12.0	−14.8	n/a	n/a
2024 (n = 13,506)	−7.5	−10.3	n/a	n/a	n/a
2025 (n = 12,508)	−6.2	n/a	n/a	n/a	n/a

t on **hindsight** ECL — valid	h1	h2	h3	h4	h5
2021	+6.1	+6.8	+7.3	+7.5	+7.6
2022	+9.4	+9.5	+9.7	+9.7	n/a
2023	+1.7	+2.4	+2.4	n/a	n/a
2024	+1.2	+1.6	n/a	n/a	n/a
2025	+3.5	n/a	n/a	n/a	n/a

Synthetic book, whole-book shadow basis. **Red** = $t > 2$ (realised above expected), **blue** = $t < -2$, green = within noise. **Reading.** Booked basis flips sign at the 2023 stress — the scenarios were wrong in opposite directions, which says nothing about the engine. Hindsight basis stays +6 to +10 for the 2021–22 cohorts (the engine under-forecast losses even with the right economy — bias confirmed), near zero for 2023–25.

F Definitions

Static cohort — every loan on book at a reference date t_o (31 Dec 2021 … 2025), frozen with the ECL it carried; loans leave only by prepayment, maturity, sale or default, none are added.

Default, τ, development year h — default = 90 days past due or unlikely to pay; year h = calendar year $t_o + h$; τ = the h in which a loan defaults; a loss belongs to its default year.

Stage 1 / 2 / 3 — 1 = performing, 12-month ECL booked (= m_1); 2 = significant increase in credit risk, lifetime ECL booked (= Σm_h); 3 = in default, ECL = EAD × in-default LGD.

Shadow lifetime ECL — lifetime ECL computed for a Stage 1 loan with the same engine — used in the test (and for survivors in the roll-forward), never booked.

Realised loss, realised LGD L — $(1+r)^{-\tau}$ × EAD(τ) × L, with $L = 1 - [\Sigma_j\, R_j(1+r)^{-(u_j-\tau)} - K]$ / EAD(τ): R_j = cash received in year u_j, K = workout cost. Zero for loans that never default.

Hindsight ECL — the same ECL engine run again at t_o on the frozen loans with the macro path that actually happened — the forecast the bank would have booked had it known the economy.